Can you guess what we would see if a four-dimensional human being living in the bulk were to pass through our brane? To speculate about this, you need to imagine what a four-dimensional human being—with two legs, a torso, two arms, and a head—must "look like" in the bulk, with its four space dimensions. And what its cross sections must look like.

The Nature of Bulk Beings, and Their Gravity



If there *are* bulk beings, what are they made of? Certainly not atom-based matter like us. Atoms have three space dimensions. They can only exist in three space dimensions, not four. And this is true of subatomic particles as well. And it is true also of electric fields and magnetic fields (Chapter 2) and the forces that hold atomic nuclei together.

Some of the world's most brilliant physicists have struggled to understand how matter and fields and forces behave if our universe really is a brane in a higher-dimensional bulk. Those struggles have pointed rather firmly to the conclusion that all the particles and all the forces and all the fields known to humans are confined to our brane, with one exception: gravity, and the warping of spacetime associated with gravity.

There might be other kinds of matter and fields and forces that have four space dimensions and reside in the bulk. But if there are, we are ignorant of their nature. We can speculate. Physicists do speculate. But we have no observational or experimental evidence to guide our speculations. In *Interstellar,* on Professor Brand's blackboard, we see him speculating (Chapter 25).

It's a reasonable, half-educated guess that, if bulk forces and fields and particles do exist, we will never be able to feel them or see them.